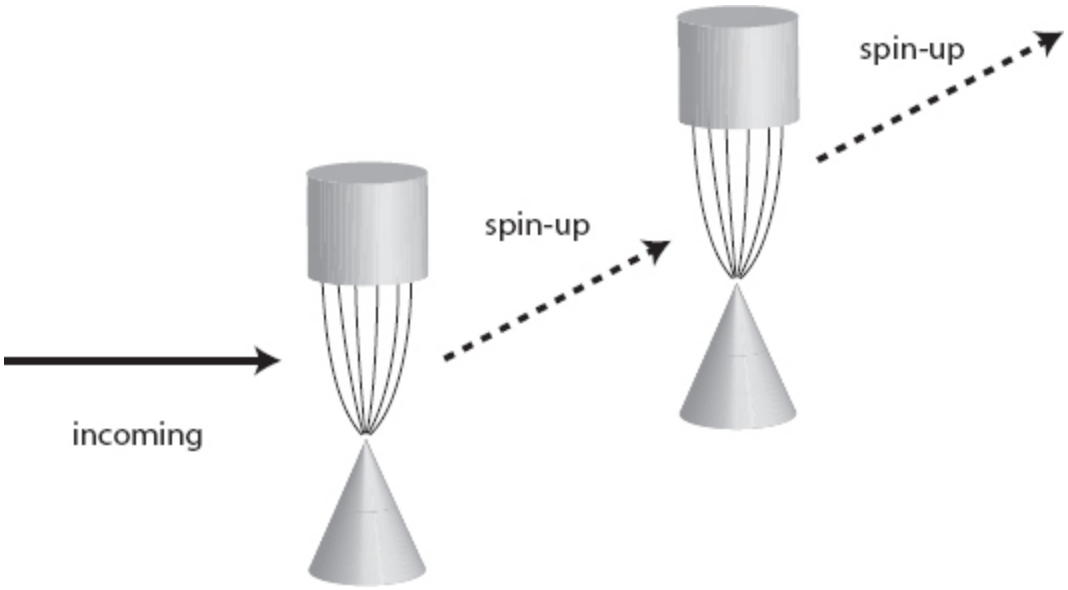
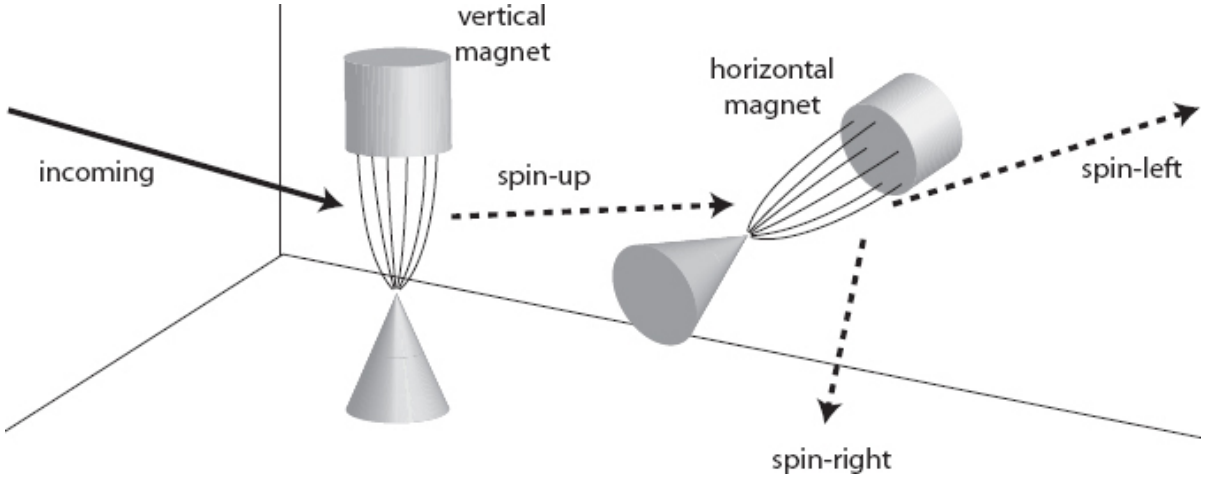


No. Instead, the second magnet splits the spin-up electrons into two beams. Half of them are deflected to the right (along the direction of the second magnet) and half of them are deflected to the left.



Madness. Our classical intuition makes us think that there is something called “the axis around which the electron is spinning,” and it makes sense (maybe) that the spin around *that* axis is quantized. But the experiments show that the axis around which the spin is quantized isn’t predetermined by the particle itself; you can choose any axis you like by rotating your magnet appropriately, and the spin will be quantized with respect to that axis.

What we’re bumping up against is another manifestation of the uncertainty principle. The lesson we learned was that “position” and